

DNS Detective (DNS Resolution Analysis)

Objective

The purpose of this study is to investigate the Domain Name System (DNS) resolution process using the nslookup command and examine how domain names are translated into IP addresses.

Tools Used

Commands used in this study:

- nslookup

Step 1 – DNS Resolution

The command was executed: nslookup google.com

Output:

Server: UnKnown

Address: 46.x.x.x

Non-authoritative answer:

Name: google.com

Addresses: 2a00:1450:4017:816::200e

172.217.16.142

* Sensitive network information has been anonymized for security purposes.

Findings:

The DNS lookup results were examined to analyze how the domain name was resolved.

- **DNS server used:** 46.x.x.x
- **DNS server name:** Unknown
- **DNS response type:** Non-authoritative answer
- Domain successfully resolved to both **IPv4** and **IPv6** addresses.

Analysis:

The nslookup results confirmed that the configured DNS server successfully resolved the domain name google.com. The response was marked as Non-authoritative, indicating that the information was retrieved from a recursive DNS server rather than directly from Google's authoritative DNS servers. The domain was resolved to both IPv4 and IPv6 addresses, demonstrating support for dual-stack networking.

Step 2 – Alternative Domain Resolution

Commands were executed:

- nslookup github.com
- nslookup openai.com

Output:

nslookup github.com

Server: UnKnown

Address: 46.x.x.x

Non-authoritative answer:

Name: github.com

Address: 140.82.121.3

nslookup openai.com

Server: UnKnown

Address: 46.x.x.x

Non-authoritative answer:

Name: openai.com

Addresses: 172.64.154.211

104.18.33.45

* Sensitive network information has been anonymized for security purposes.

Analysis:

Additional DNS queries were performed using different domain names to compare DNS responses. Although all queries were processed by the same configured recursive DNS server and returned Non-authoritative responses, the results differed according to each domain's DNS configuration. github.com resolved to a single IPv4 address, while openai.com resolved to multiple IPv4 addresses. This demonstrates that different services may use different DNS configurations depending on their infrastructure, load balancing, and traffic management strategies.

Step 3 – Reverse DNS Lookup

The command was executed: nslookup 8.8.8.8

Output:

Server: UnKnown

Address: 46.x.x.x

Name: dns.google

Address: 8.8.8.8

* Sensitive network information has been anonymized for security purposes.

Findings:

The reverse DNS lookup results were examined to identify the domain name associated with a public IP address.

- **Configured DNS server:** 46.x.x.x
- **Reverse lookup target:** 8.8.8.8
- **Resolved domain name:** dns.google
- Reverse DNS lookup completed successfully.

Analysis:

The reverse DNS lookup successfully resolved the public IP address 8.8.8.8 to the domain name dns.google. This demonstrates that reverse DNS records (PTR records) can be used to map IP addresses back to their associated hostnames. Unlike forward DNS resolution, which translates domain names into IP addresses, reverse DNS performs the opposite operation.

Key Concepts Learned

During this exercise, the following concepts were reinforced:

- Forward DNS lookup
- DNS resolution process
- Recursive DNS server
- Non-authoritative DNS response
- Domain name to IP address translation
- IPv4 and IPv6 addressing
- Multiple A records (load balancing)
- Reverse DNS lookup (PTR records)

Conclusion

The DNS analysis successfully demonstrated how domain names are translated into IP addresses using the Domain Name System. The study also showed that different domains may resolve to different numbers and types of IP addresses depending on their DNS configuration. In addition, the reverse DNS lookup confirmed that IP addresses can be mapped back to hostnames through PTR records. Overall, this lab provided practical experience in understanding and analyzing DNS resolution, an essential component of modern computer networks and network troubleshooting.